HERIOT WATT UNIVERSITY



B37VR ROBOTICS PROJECT

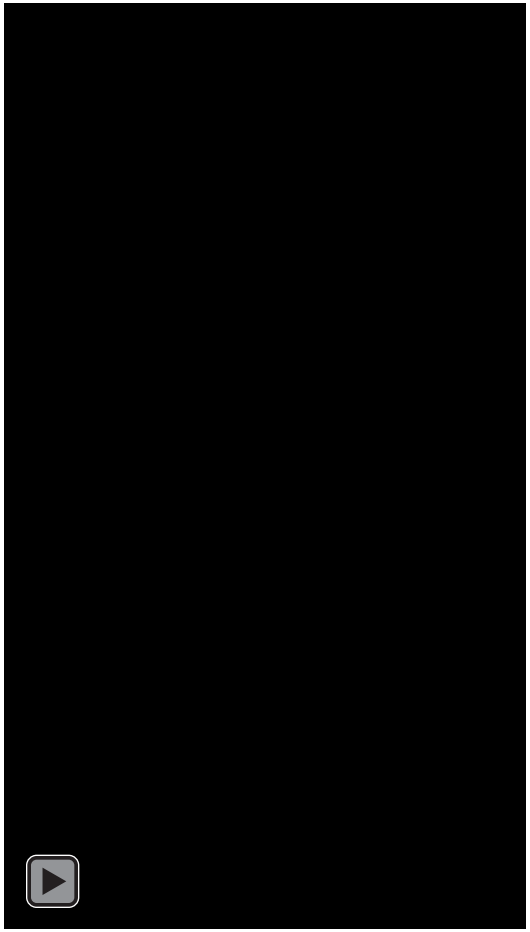
4 Setting up the robot to follow light and looking at the IR sensors for line following

January '21

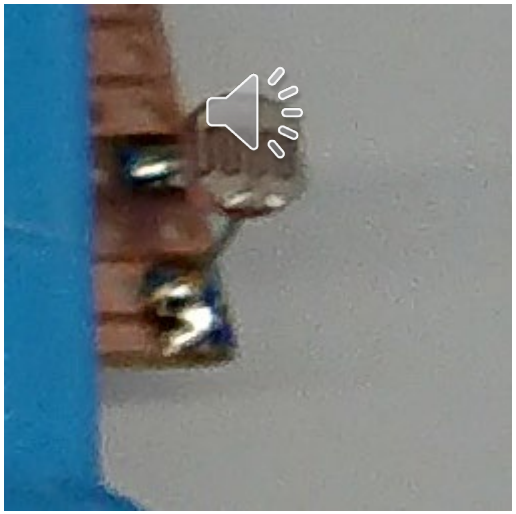
4 SUMMARY

- **LEARNING RESOURCES AVAILABLE**
  - Robotic buggies
  - LDR and resistors for voltage divider
  - Code for light following
  - IR sensors, resistors and strip boards for constructing line following sensor
- **LEARNING ACTIVITIES**
  - Try to get the robot to follow light by connecting in sensors and editing code
- **TASKS TO BE COMPLETED**
  - Get robot to follow light source
  - Construct line following sensor with 3 IR sensors
  - Characterise IR sensors for use in line following sensor

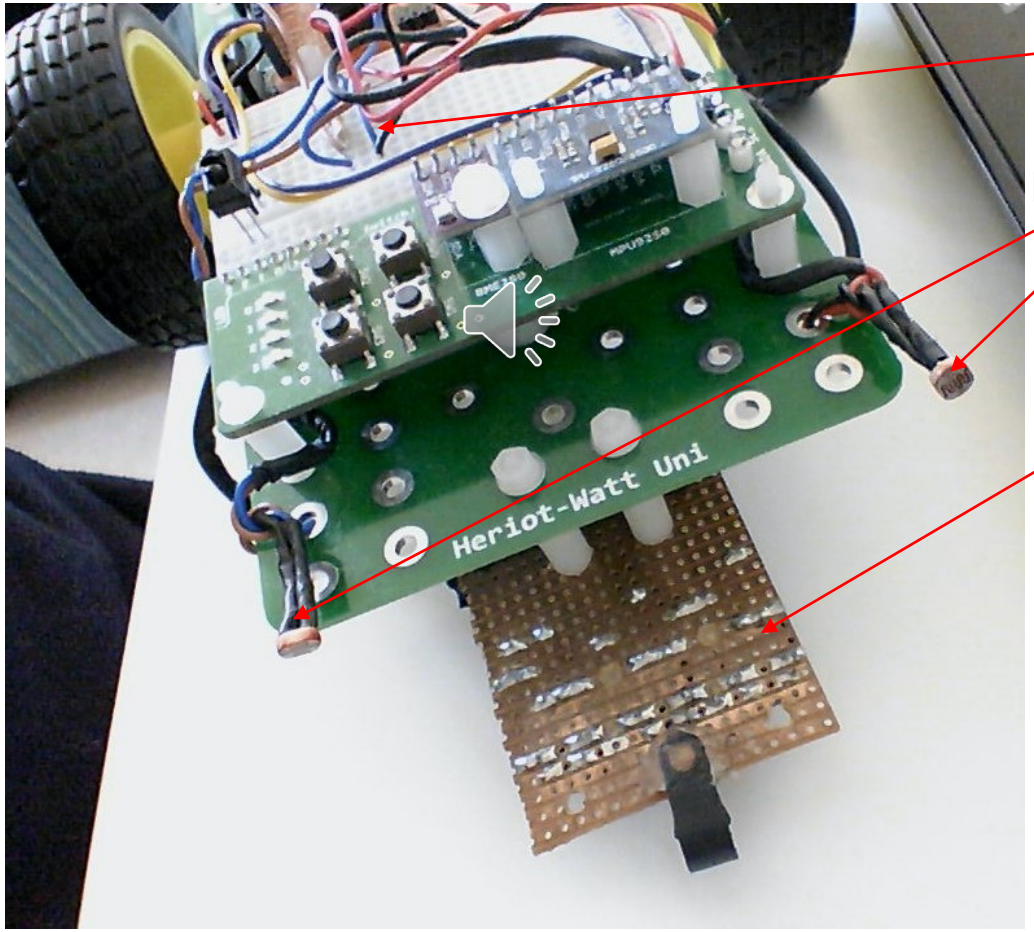
ROBOT FOLLOWING A TORCH LIGHT



AN EXAMPLE OF MOUNTING LDR FOR LIGHT FOLLOWING



ANOTHER EXAMPLE OF MOUNTING LDRS FOR LIGHT FOLLOWING



Connection to resistors for voltage divider

LDRs on each side for light following

Board for IR sensors for line following

LIGHT FOLLOWING CODE PART 1 SETTING UP

```
/*
Light following control of two Bi-directional Motors
*/

int enablePin1 = 9;
int enablePin2 = 10;
int in1Pin = 2;
int in2Pin = 3;
int in3Pin = 4;
int in4Pin = 5;
int sensDiffThresh = 50;
int SensorLeft = 0;
int SensorRight = 0;
/* Define the pins for the light sensors */
int LeftSensor = A1; // This stores the value of the Left Sensor pin to use later on in the sketch
int RightSensor = A2; // This stores the value of the Right Sensor pin to use later on in the sketch
const int slowSpeed = 100; // the range for speed is(0,255)
const int maxSpeed = 250; // the range for speed is(0,255)

/* Define variables for the LDR Sensor readings and the difference between the two */
int SensorDifference = 0; // This value is used to determine the difference between the Left and Right
int error = 40;// difference between motors for turning

/* variables to keep track of current speed of motors */
int leftServoSpeed = 0;
int rightServoSpeed = 0;

void setup()
{
  pinMode(in1Pin, OUTPUT);
  pinMode(in2Pin, OUTPUT);
  pinMode(in3Pin, OUTPUT);
  pinMode(in4Pin, OUTPUT);
  pinMode(enablePin1, OUTPUT);
  pinMode(enablePin2, OUTPUT);

}
```

LIGHT FOLLOWING CODE PART 2 SENSOR READING IN MAIN LOOP

```
void loop()
{
 boolean reverse = 0; //digitalRead(switchPin);
SensorLeft = 1023 - analogRead(LeftSensor); // This reads the value of the sensor, then saves it to the corresponding integer.

delay(1); // delay to let adc settle

SensorRight = 1023 - analogRead(RightSensor); // This reads the value of the sensor, then saves it to the corresponding
integer.

delay(1); // delay to let adc settle

SensorDifference = abs(SensorLeft - SensorRight); // This calculates the difference between the two sensors and then saves it
to an integer.
```

LIGHT FOLLOWING CODE PART 3 DETERMINING WHAT SPEED MOTORS SHOULD RUN AT

```
// This section of the sketch is what actually interprets the data and then runs the motors accordingly.

if (SensorLeft > SensorRight && SensorDifference > sensDiffThresh) { // This is interpreted as if the Left sensor reads more
light than the Right Sensor, Do this:
leftServoSpeed = slowSpeed; // This is used to turn left making the left servo much slower than the right servo
rightServoSpeed = maxSpeed; // opposite motor runs to turn Left
}

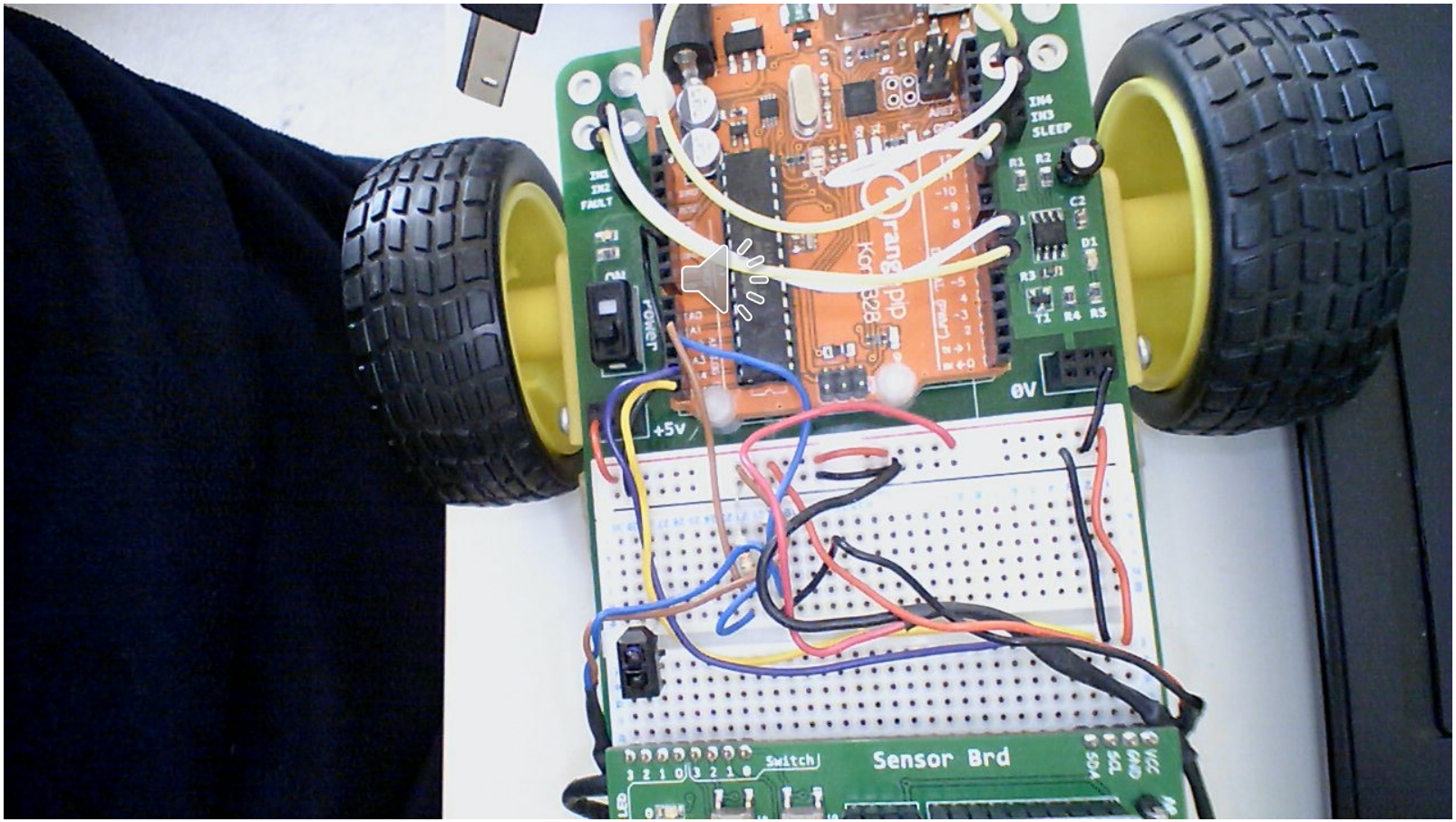
if (SensorLeft < SensorRight && SensorDifference > sensDiffThresh) { // This is interpreted as if the Left sensor reads less
light than the Right Sensor, Do this:
leftServoSpeed = maxSpeed; // This is used to go right
rightServoSpeed = slowSpeed; // This is used to turn right making the left servo much slower than the left servo.
}

else if (SensorDifference < sensDiffThresh) { // This is interpreted as if the difference between the two sensors is less
than the threshold
                    //(Experiment to suit our sensors), Do this:
leftServoSpeed = maxSpeed; // This is used to go straight
rightServoSpeed = maxSpeed; // both motors are enabled to go forward.
}
  setMotor(leftServoSpeed, rightServoSpeed, reverse);
}
```

LIGHT FOLLOWING PART 4 CODE TO SEND MOTOR SPEEDS

```
void setMotor(int leftServoSpeed, int rightServoSpeed, boolean reverse)
{
  analogWrite(enablePin1, rightServoSpeed); //pwm the enable pin with the speed value for the right motor
  digitalWrite(in1Pin, ! reverse);
  digitalWrite(in2Pin, reverse);
  analogWrite(enablePin2, leftServoSpeed); //pwm the enable pin with the speed value for the left motor
  digitalWrite(in3Pin, ! reverse);
  digitalWrite(in4Pin, reverse);
}
```

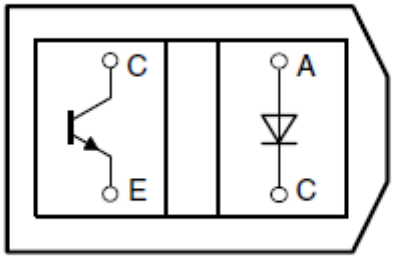
CONNECTIONS OF SENSORS TO MICROCONTROLLER BOARD AND FROM CONTROLLER BOARD TO H-BRIDGE



IR SENSOR



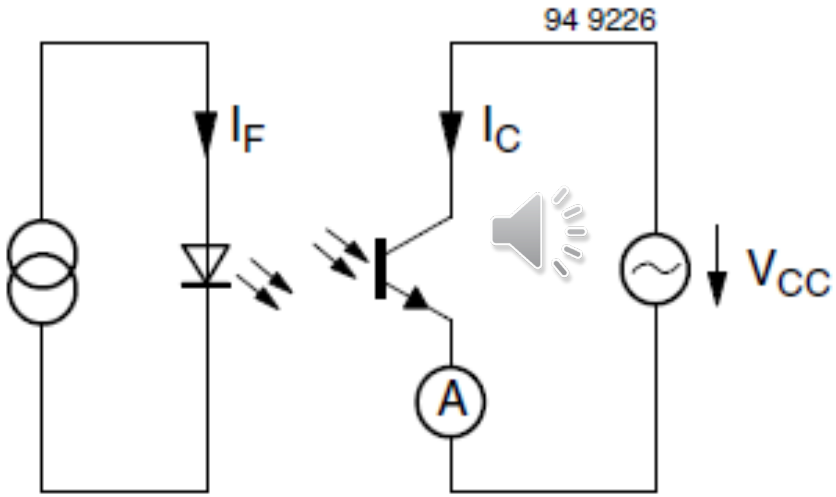
19156_2



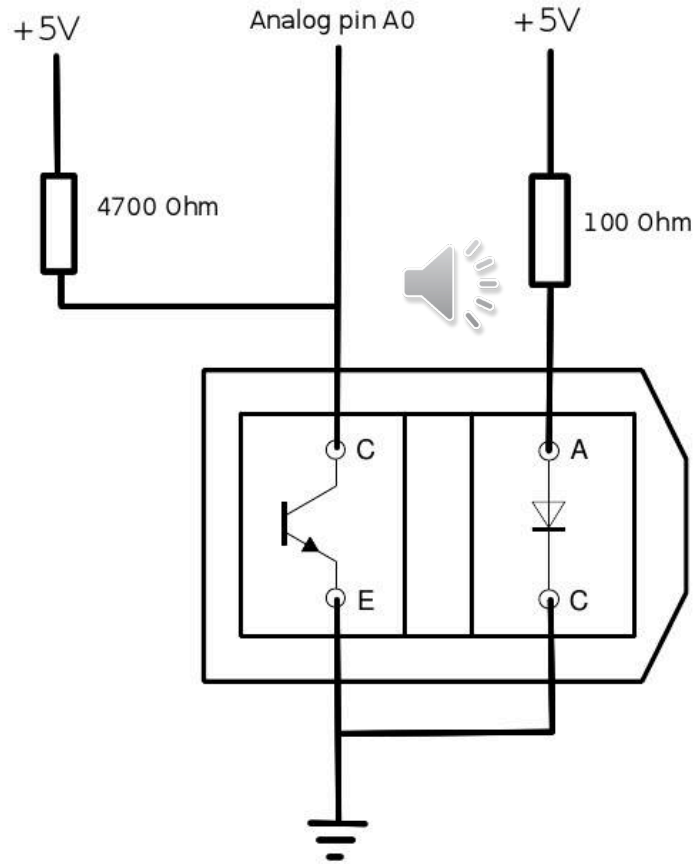
Top view

19156_1

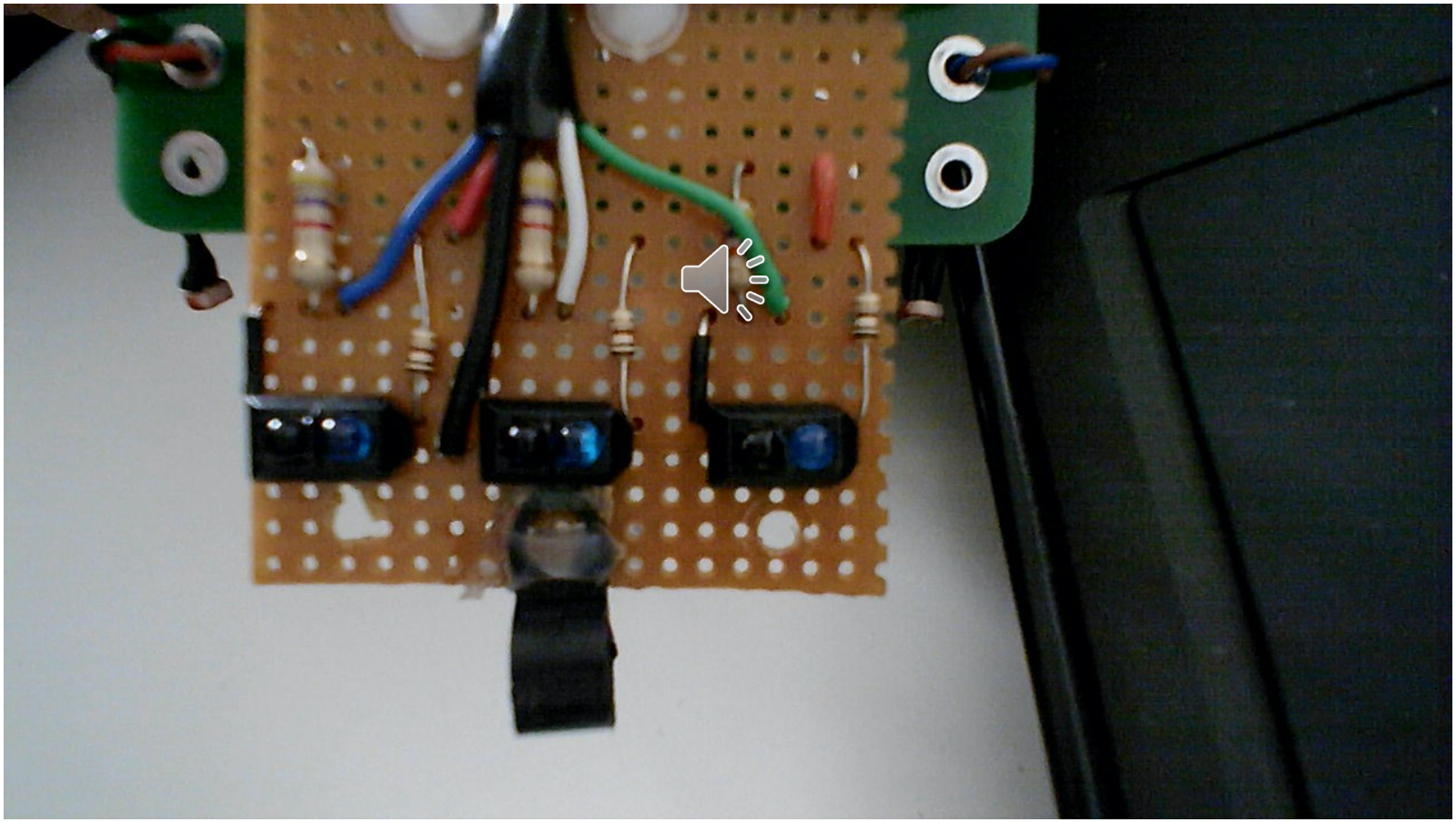
BASIC OPERATION OF IR SENSOR



CIRCUIT FOR IR SENSOR



LAYOUT OF 3 IR SENSORS FOR LINE FOLLOWING



CONSTRUCT SENSOR BOARD

- First layout your components on the strip board to see how it will fit
- Cut stripboard to size. Remember to leave enough space on the board to fit mounting screws. See document on using strip board for circuits if you are not familiar with using strip board
- Cut the tracks in the appropriate places. Add you components and connection wires and solder you sensor board
- When the sensor board is complete undertake the sensor characterisation

CODE FOR IR SENSOR CHARACTERISATION

```
/*
Read 3 IR sensors for callibration
  Reads an analog input on pin 3,4 &5, prints the result to the Serial Monitor.
  Graphical representation is available using Serial Plotter (Tools > Serial Plotter menu).
*/

// the setup routine runs once when you press reset:
void setup() {
  // initialize serial communication at 9600 bits per second:
  Serial.begin(9600);
}

// the loop routine runs over and over again forever:
void loop() {
  // read the input on analog pin 0:
  int sensorValueLeft = analogRead(A5);
  delay(2);
   int sensorValueCentre = analogRead(A4);
  delay(2);
   int sensorValueRight = analogRead(A3);
  delay(2);
  // print out the value you read:
  Serial.print("left=");
  Serial.print(sensorValueLeft);
   Serial.print("centre=");
  Serial.print(sensorValueCentre);
   Serial.print("right=");
  Serial.println(sensorValueRight);
  delay(2000);        // delay in between reads
}
```

TASKS TO BE COMPLETED

- Get robot to follow light source
- Construct line following sensor with the 3 IR sensors
- Characterise IR sensors for use in line following sensor

